

\$1,800 PC Build

My PC: <https://pcpartpicker.com/list/L76H9K>

Because I built my PC with PC Part Picker, I know everything should be compatible because the site has a system that checks for compatibility, and it raised no errors for my build.

CPU: [AMD Ryzen 7 9800X3D 4.7 GHz 8-Core Processor](#) (\$439.99)

I picked this CPU because it's very powerful both for general productivity and gaming purposes, and I feel it's appropriate for this budget. What I like most about it is that it has a base clock rate of 4.7GHz, and it gets up to 5.6 GHz overclocked which is great for gaming. Another upside of this CPU is that the GPU I chose is also AMD, which means that my PC can use AMD's Smart Access Memory, a feature that allows the CPU to have full access to the GPU frame buffer, giving me a slight boost in performance. At idle, the CPU draws about 96 watts of power, which is less than the Intel i9-14900K, its Intel equivalent.

CPU Cooler: [ARCTIC Liquid Freezer III Pro 360 77 CFM Liquid CPU Cooler](#) (\$89.99)

Since the Ryzen 7 9800X3D only pulls around 96 watts idling, this cooler is more than enough to sustain a comfortable temperature, even in heavy load situations. It has a 360 mm radiator which is larger than most all-in-one coolers for this price range. With it being an all-in-one, it's simple to install and sustains CPU temperatures well within the AMD recommendations. Something else I like about this cooler is that it features a VRM fan, which helps cool the voltage regulator module on the motherboard.

Motherboard: [Gigabyte B850 AORUS ELITE WIFI7 ATX AM5 Motherboard](#) (\$199.99)

I went with this motherboard because it's compatible with the CPU I chose (AM5 support), and it has ample power delivery for my components, especially the GPU. If I were to actually get this PC, I would overclock the CPU, and this motherboard's great power delivery is perfect for that case in my opinion. Another big reason I chose this board is that it supports DDR5 RAM which is very important for me. Not only is the difference between DDR4 and DDR5 very noticeable, but the CPU I chose only supports DDR5, making this board the clear pick.

Storage: [Samsung 990 EVO PLUS SSD 1TB](#) (\$164.99)

After initially choosing the Kingston NV3 1 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive, I switched to this Samsung 990 Evo Plus because when I did some research, I found out that the Kingston NV3 have some issues (it's a budget buy), like one-sided overheating, slow random access speeds, and slow large file transfers. I realized that those issues would make windows feel slower in general, not to mention a less than ideal gaming experience. The SSD I went with should give me a quicker and more responsive feel in general, especially with large file transfers and level loading in games.

GPU: [Gigabyte GAMING OC Radeon RX 9060 XT 16 GB Video Card](#) (\$459.95)

Like I mentioned, having an AMD CPU and GPU unlocks Smart Access Memory (SAM). This is especially important for this GPU because it has a relatively large VRAM area (16 GBs). With SAM, this should give me a noticeable buff in performance. The closest Nvidia equivalent to this GPU is the RTX 5060 Ti, which is a good mid-range GPU that can handle most games, especially with the CPU I picked. Another reason I like AMD GPUs is that it has a feature called HYPR-RX, which AMD claims boosts FPS by up to 3x. The cooling for this specific model is also good because it utilizes three fans for air flow. As for the power, this GPU draws approximately 160 watts, making it the most power-hungry component in this build.

Power Supply: [MSI MAG A850GL PCIE5 850 W 80+ Gold Certified Fully Modular ATX Power Supply](#) (\$117.99)

This 850 watt power supply is plenty for the components I picked, even with an overclocked CPU. With a full load peak of all of my components (~410W), it wouldn't come close to the maximum wattage for the PSU. A transient spike could potentially reach 600W, but even that isn't beyond the PSU's limit (although it could damage components in a worse-case scenario).

Case: [Lian Li O11 VISION COMPACT ATX Mid Tower Case](#) (\$124.99)

The GPU I picked is quite big because of its 3-fan cooling system, so it requires a large case to fit. This case's dimensions are 17.6 x 11.3 x 17.57 inches, which is compatible with all of my components. The case is specifically designed for ATX motherboards, which is perfect for the motherboard I chose. It also has good PSU clearance, which is necessary since the 850W PSU I chose is bigger than average. The CPU cooler I chose is a high-end cooler that reaches peak cooling performance with good airflow, which this case provides.

RAM : 2x [Kingston FURY Beast 16 GB](#) (32GB Total)

32 GBs of RAM for this gaming build is great in my opinion. It would give me a balance of performance and speed, especially considering it's 32 GBs of DDR5. The reason I split the 32 GBs into two 16 GB sticks is because utilizing dual channel memory increases bandwidth considerably. AMD provides a feature called EXPO (extended profiles for overclocking) that this RAM set is compatible with. EXPO allows users to change the profile of their memory (raise max speed). In other words, it's an easy way to overclock your RAM so that you get more bang for your buck.

Conclusion

Total (not including RAM) : \$1715.88

Overall, this turned out to be a higher-end mid-range build that's perfect for gaming and general productivity. My favorite features were the DDR5 RAM space and overclocked rate of the CPU. Another aspect of this build that I felt is worth mentioning is that the compatibility between the AMD CPU and AMD GPU allow for many AMD build-in firmware and software features that generally extend the performance of the two components.

Both of those components combined should give the user a really responsive experience when using the computer for non-gaming purposes. If I were to pick a display for this build, I would definitely go with a 1440p minimum at around 240Hz so that I can experience the full potential of the overclocked CPU and GPU.

As for gaming, the GPU and CPU combination is well-suited for 200+ FPS on 1080p , and ~100 FPS on 1440p for most new games and almost all other games. Seeing as how this is a PC specifically for gaming, I'm sure it would make for an excellent gaming experience and last for a long time with proper maintenance.

The total price is fair for this build, but it would be higher if I were to include the RAM prices at the non-inflated rate. Generally, storage hardware is extremely inflated at the moment (even SSDs) because of the AI boom, but as that settles down, the price should land somewhere around the \$1850-\$1900 mark.

The biggest help for me in building this PC was the website I used: PC Part Picker. They have a useful system that checks for part compatibility that works very well generally, although it's not perfect in some edge cases. Another feature I love about this site is that they give you the best price from merchants for each part, which helped me immensely in staying within budget.

